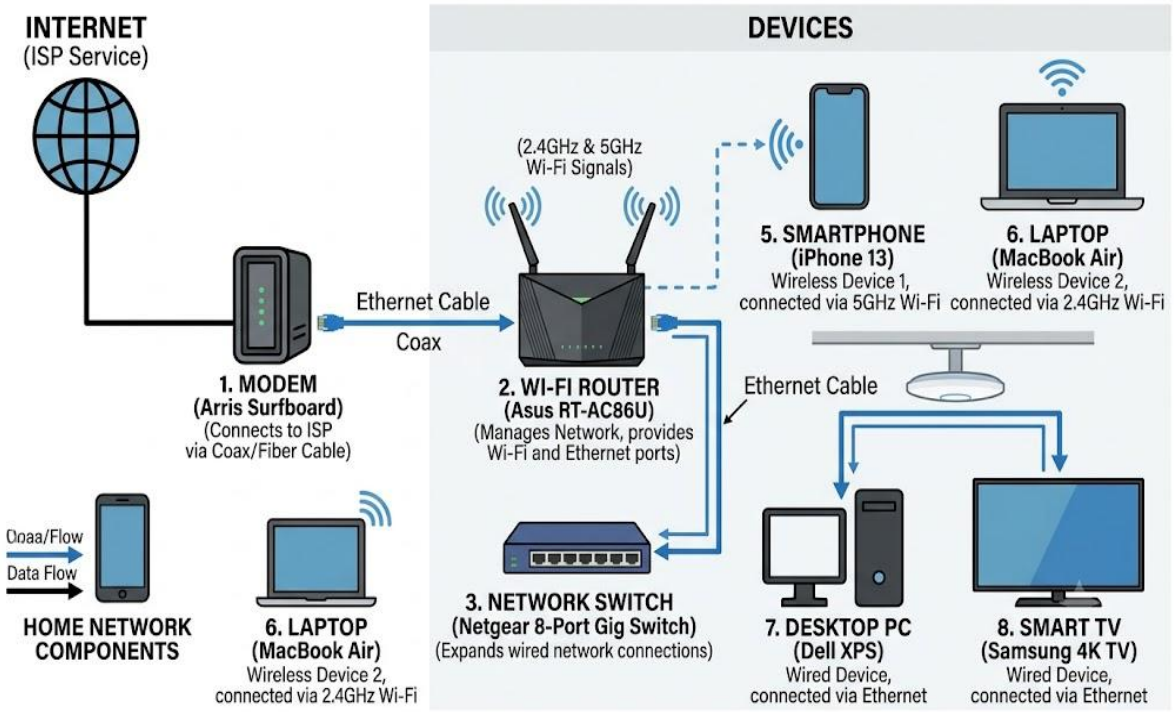
1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2.List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Network Hardware Overview

- **Modem**
    - **Main Function:** Translates incoming signals from your Internet Service Provider (ISP) into digital data your home network can process, and vice versa.
    - **Real-Life Example:** Sitting next to your cable outlet, connecting your house directly to your internet provider's fiber, coax, or phone line.
- **Router**
    - **Main Function:** Directs network traffic between the internet and your local devices, assigning unique local IP addresses and managing data flow.
    - **Real-Life Example:** Sitting on a desk at home, allowing multiple family members to share a single internet connection at the same time.
- **Switch**

- **Main Function:** Connects multiple wired devices within the same local network, allowing them to communicate directly with each other at high speeds.
- **Real-Life Example:** Mounted in a server rack at an office to connect dozens of desktop computers and printers to the local network using physical cables.

- **Access Point (AP)**

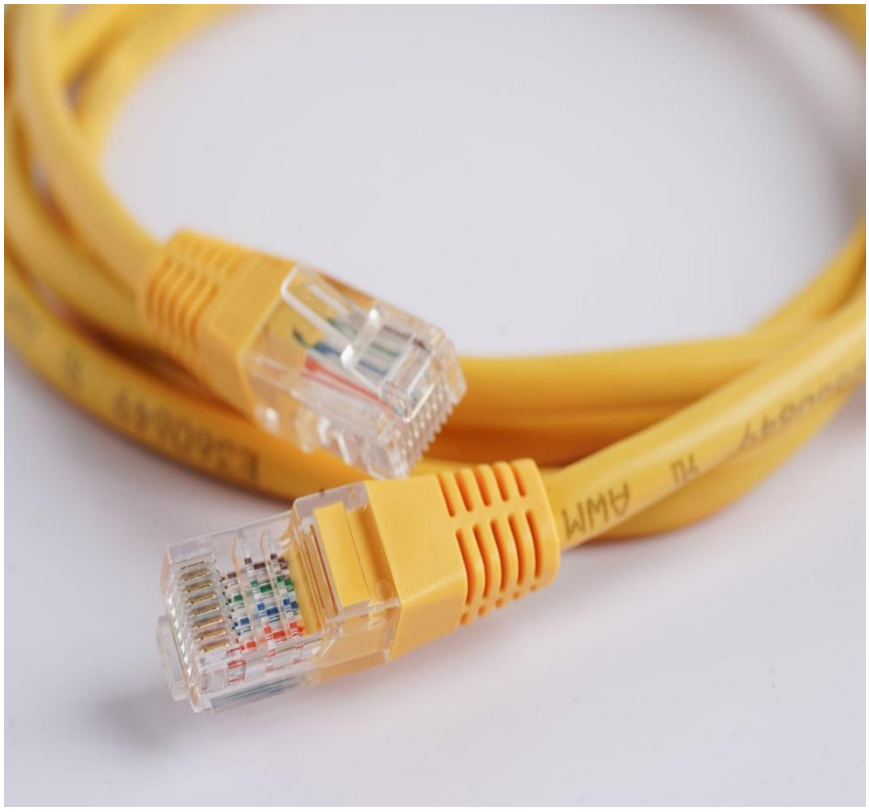
- **Main Function:** Transmits wireless signals (Wi-Fi) so wireless devices can connect to a wired network.
- **Real-Life Example:** Mounted on the ceiling of a coffee shop, hotel hallway, or school room to extend Wi-Fi coverage across a large area.

- **Network Cable (Ethernet Cable)**

- **Main Function:** Provides a physical medium for high-speed, secure, and low-latency data transmission between hardware devices.
- **Real-Life Example:** Plugged into the back of a gaming console or desktop PC to ensure a fast, stable connection without Wi-Fi lag.

3.Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet Cable (Twisted Pair / RJ45):



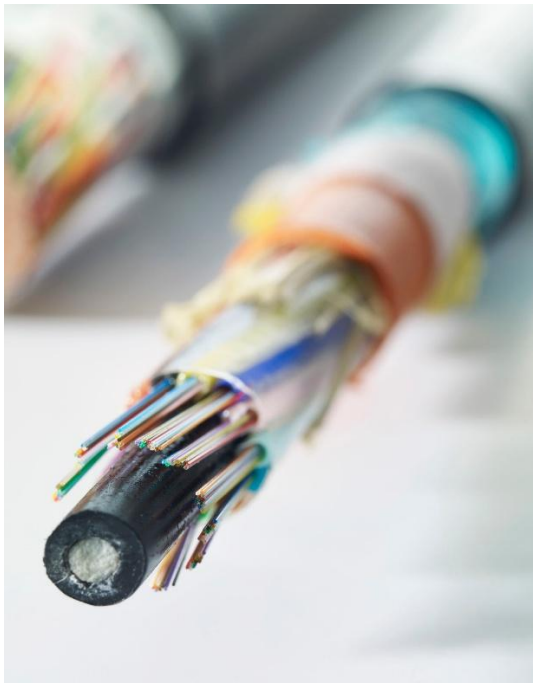
Common Use: Used to connect computers, routers, gaming consoles, and switches within Local Area Networks (LANs) for high-speed local internet access.

2. Coaxial Cable:



Common Use: Used primarily to transmit cable television signals, connect satellite dish antennas, and attach cable modems to wall outlets.

3. Fiber Optic Cable



Common Use: Used for long-distance, high-bandwidth data transmission, such as telecommunication backbones, submarine internet cables, and FTTH (Fiber to the Home) broadband connections.

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data

Feature	Router	Switch
Main Use	Connects multiple distinct networks together and routes traffic between them (e.g., connecting a local network to the internet).	Connects multiple devices together within a single Local Area Network (LAN) to allow device-to-device communication.
Typical Location	Situated at the boundary/gateway of a network (between your local network and the Internet Service Provider).	Situated inside the local network, connecting end devices like computers, printers, and servers.
How Data is Handled	Analyzes IP addresses (Layer 3 - Network Layer) to determine the best path for sending data packets across different networks.	Reads MAC addresses (Layer 2 - Data Link Layer) to forward data frames directly to the intended destination device on the local network.
Data Unit Processed	Packets	Frames